AKHIL ADDAPA

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PROFESSIONAL SUMMARY

- Software Engineer with 4+ years of hands-on experience delivering production-grade solutions across fintech, SaaS, and real-time data platforms. Specialized in building scalable, event-driven microservices with Java, Spring Boot, Kafka, and React. Proven ability to lead end-to-end development — from designing APIs and automating CI/CD pipelines to optimizing cloud deployments and resolving live production issues. Adept at cross-functional collaboration, agile delivery, and driving business value through robust software solutions.

TECHNICAL SKILLS

Programming Languages: Java, Python, C, JavaScript (ES6), TypeScript, SQL
Web & Front-End Development: HTML, CSS, React.js, Angular, Babel, jQuery
Backend & System Design: Spring Boot, RESTful APIs, GraphQL, Apache Kafka, Tomcat, Microservice Architecture
Cloud & DevOps: AWS (EC2, S3, ECS, Lambda, CloudWatch, IAM), Docker, Kubernetes, Jenkins, CI/CD Pipelines, Maven, Gradle
Databases & Caching: PostgreSQL, Oracle, MySQL, MongoDB, Cassandra, Hibernate, Redis
Monitoring, Testing & Tools: Splunk, Postman, IntelliJ IDEA, Eclipse, Visual Studio Code, Git, GitHub, JUnit, Mockito, Cucumber, SonarQube

PROFESSIONAL EXPERIENCE

Capital One — *Feb 2024 – Present*
Senior Software Developer — *Mclean, VA*

- Designed and implemented distributed microservices using Java 21 and Spring Boot 3, contributing to a scalable financial dispute resolution system capable of processing thousands of transactions per second with minimal latency.
- Spearheaded the integration of card networks into existing backend flows, automating dispute workflows that previously required manual reconciliation by operations teams.
- Enhanced customer servicing APIs to support real-time status tracking and asynchronous updates using Kafka event streams and consumer groups, improving transparency in dispute handling.
- Collaborated with DevOps and Security to create Dockerized deployments on ECS Fargate with zero-downtime rollout strategies using blue/green and canary release patterns.
- Defined and enforced custom retry policies, circuit breakers (via Resilience4j), and timeout configurations to improve resiliency in high-throughput services dealing with third-party APIs.
- Migrated legacy synchronous endpoints to asynchronous messaging flows using Kafka producers and consumers, reducing end-to-end processing time and improving overall system decoupling.
- Developed multiple Splunk dashboards for observability, including application health metrics, error logs, and latency visualizations, allowing for early anomaly detection and reducing mean time to recovery.
- Led backend design of a feature flag system for selectively enabling card network integrations, enabling phased rollouts and reducing risk during production deployments.
- Contributed to the modernization of Angular-based internal tools by adding reusable components, responsive layouts, and enhanced form validation, which improved system usability for internal operations users.
- Worked closely with the product team to gather complex functional requirements for new dispute reason codes and translated them into elegant technical designs adhering to Capital One's domain-driven design principles.
- Implemented authentication and authorization using Spring Security with JWT, role-based access control, and integration with internal OAuth2 services to ensure secure access across distributed components.
- Introduced a Kafka DLQ (dead letter queue) monitoring system to track failed message processing events, enabling support teams to react quickly to anomalies and minimize customer impact.

- Regularly participated in production on-call rotations, using tools like CloudWatch, Splunk, and PagerDuty to diagnose and resolve live production issues across multiple microservices.
- Collaborated with QA and SRE teams to establish contract testing using WireMock and OpenAPI specifications, ensuring backward compatibility and minimizing service integration failures.
- Designed database schemas and optimized PostgreSQL and Oracle queries for critical transaction tables to ensure ACID compliance and optimal indexing strategies for low-latency access.
- Implemented robust unit and integration testing strategies using JUnit 5 and Mockito, achieving high test coverage and reducing bugs introduced during sprint deliveries.
- Actively contributed to knowledge sharing by hosting internal "tech deep dive" sessions on topics like microservice resilience patterns, Kafka performance tuning, and best practices in API versioning.
- Played a key role in architectural discussions and reviews, evaluating proposals related to CQRS, event sourcing, and data partitioning strategies for distributed domains in the dispute processing space.
- Championed Agile practices in the team by promoting peer reviews, refining user stories in collaboration with product owners, and taking ownership of sprint demo presentations for senior leadership.

Squad Software Inc. *Jul 2023 – Jan 2024*
Software Developer *Iselin, NJ*

- Designed and implemented robust, scalable REST APIs using Spring Boot and Spring Cloud, serving both internal tools and enterprise client portals that handled diverse consulting use cases.
- Built and maintained reusable frontend components using React.js and TypeScript, improving modularity and reducing frontend code duplication across multiple projects.
- Developed and secured JWT-based authentication with access and refresh token logic, enabling secure user sessions and permission enforcement across backend APIs.
- Refactored legacy backend logic and introduced DTO (Data Transfer Object) layers and service abstraction, significantly improving code readability and maintainability.
- Optimized complex queries by tuning PostgreSQL joins, leveraging indexes, and restructuring transactional boundaries using Hibernate and JPA.
- Introduced Redis-based caching mechanisms for high-frequency endpoints to reduce database load and enhance response times for end users.
- Built and deployed containerized applications on AWS ECS, creating reproducible environments using Dockerfiles and automating deployment with GitHub Actions.
- Integrated AWS CloudWatch alarms and dashboards to monitor production services, enabling rapid response to high-latency or failed requests.
- Wrote unit, integration, and contract tests using JUnit, Mockito, and TestContainers, achieving strong coverage and enabling safe refactoring during sprints.
- Developed and documented OpenAPI (Swagger) specs for all APIs, improving clarity for frontend developers and third-party integrators.
- Actively contributed to frontend performance optimization by implementing code splitting, lazy loading, and efficient Redux state management strategies.
- Introduced Role-Based Access Control (RBAC) in the backend to enforce granular access based on user roles, ensuring data security and audit traceability.
- Played a key role in feature planning and sprint estimation, ensuring technical feasibility aligned with product timelines and client expectations.
- Facilitated peer code reviews, sharing architecture suggestions, identifying potential bugs, and driving the team toward consistent clean code practices.
- Conducted knowledge-sharing sessions on API design best practices, secure token management, and efficient use of HTTP caching headers.
- Developed and maintained CI/CD pipelines using Jenkins, enabling build automation, static code analysis (SonarQube), and test execution pipelines.
- Created shell scripts to manage backups, database snapshots, and service restarts for QA environments to support testing without delays.

- Led debugging sessions across teams when performance regressions or service outages occurred, using Splunk, Postman, and VisualVM to isolate root causes.
- Collaborated directly with business stakeholders to translate user stories and requirements into actionable, modular backend components.
- Designed and managed database schema evolution using Flyway, ensuring safe and trackable migrations across staging and production environments.

App Orchid Inc. *Apr 2021 – Nov 2021*
Software Engineer *Hyderabad, India*

- Developed and maintained RESTful APIs using Spring Boot for a Natural Language Processing (NLP)-powered analytics platform tailored for the energy and utilities sector, supporting real-time data ingestion and dashboard rendering.
- Designed and optimized MongoDB document schemas to support high-velocity sensor data storage and enable fast, efficient querying for analytics dashboards.
- Collaborated with UI/UX designers and frontend developers to build and enhance Angular-based modules, improving client interaction workflows and minimizing form errors through reactive validation.
- Played a key role in transitioning the application to microservices architecture, breaking down monolith components and defining contracts between services via REST endpoints.
- Conducted performance tuning of backend services, leveraging Java profilers and MongoDB query analysis to reduce API response time and memory overhead.
- Wrote detailed unit and integration tests using JUnit, Mockito, and TestRestTemplate to validate business logic and service-to-service communication.
- Created Postman collections and API test cases to validate critical application flows, which were used by QA teams and during UAT with enterprise customers.
- Implemented secure user authentication and role-based authorization using Spring Security and JWT, enabling fine-grained access controls and tenant isolation.
- Participated in Agile sprints, collaborating with cross-functional teams during planning, grooming, and retrospective sessions to improve process efficiency and output quality.
- Integrated third-party APIs (weather, regulatory data, utility rates) to enrich platform insights and extend analytics capabilities for energy forecasts and anomaly detection.
- Worked closely with QA to define and execute regression testing plans, helping to reduce production bugs and maintain high release stability.
- Created UI/UX audit reports to highlight inconsistencies and bottlenecks, working with the product team to design data-driven improvements.
- Developed API usage logging and audit trails to support enterprise compliance, security, and debugging needs across multiple modules.
- Created backend modules for customer segmentation, anomaly detection, and alert generation based on predefined business logic rules.
- Used Git branching strategies and merge request processes to maintain a clean codebase, improve team collaboration, and streamline release processes.
- Assisted DevOps in writing CI build jobs and deployment pipelines for staging environments using Maven, Jenkins, and shell scripting.
- Built custom error handlers and standardized API error responses across microservices to improve debuggability and maintain consistency.
- Documented module-level system architecture, API workflows, and key design decisions for future onboarding and maintenance.
- Collaborated with business analysts and data scientists to shape backend services that powered advanced energy usage forecasting algorithms.

Apisero Global Integrations Pvt. Ltd. *May 2020 – Mar 2021*
Software Engineer *Hyderabad, India*

- Developed MuleSoft-based integration APIs for enterprise clients in insurance, finance, and retail sectors, enabling seamless data flow between legacy systems, SaaS platforms, and cloud-native applications.
- Designed layered API structures (Experience, Process, System) following MuleSoft's API-led connectivity model, improving modularity and reusability across integration projects.
- Utilized DataWeave to perform complex transformations between JSON, XML, CSV, and SOAP payloads, supporting diverse client-specific formats and protocols.
- Built and configured MuleSoft connectors for Salesforce, SAP, and Oracle to streamline business workflows such as lead creation, invoicing, and real-time order processing.
- Collaborated with cross-functional teams to capture integration requirements and convert them into working prototypes, later scaling them into production-grade APIs.
- Authored custom error handling strategies and reusable exception handler flows that provided consistent messaging and fallback options for all integrations.
- Wrote PL/SQL stored procedures and triggers in Oracle to automate backend validation and batch updates for data pipeline workflows.
- Led efforts to optimize batch data processing jobs, reducing execution time by identifying memory leaks and tuning heap size allocations within Mule runtime.
- Designed and executed integration performance benchmarks, including load testing and throughput monitoring, to ensure SLAs were met under peak usage conditions.
- Assisted in writing CI/CD pipelines using Jenkins, automating builds, deployments, and environment-specific configuration management for MuleSoft projects.
- Utilized Anypoint Studio to build, test, and debug integration flows, ensuring smooth delivery and timely issue resolution during all stages of the SDLC.
- Participated in client meetings to gather and validate mapping rules and field-level transformations during early discovery and planning phases.
- Created and maintained RAML specifications and published API documentation using Anypoint Exchange to support reuse and governance.
- Integrated with external systems using various transport protocols such as HTTPS, FTP, JDBC, JMS, and custom connectors for legacy endpoints.
- Assisted in designing event-driven data pipelines that responded to upstream changes, triggering real-time synchronization across services and avoiding polling inefficiencies.
- Delivered key modules ahead of deadlines for time-sensitive projects by leveraging reusable components and parallel development strategies.
- Collaborated with QA and UAT teams to create test scenarios, mock payloads, and edge case coverage matrices that ensured production readiness.
- Wrote scripts for data validation and backfilling during cutover periods, minimizing downtime during go-live transitions.
- Monitored integration flows and implemented log aggregation strategies for better traceability and proactive error detection using custom log formats.
- Actively contributed to internal knowledge base and onboarding playbooks by documenting reusable templates, connector behaviors, and deployment guides.

EDUCATION

Kennesaw State University, May 2023 — Georgia, USA
Master of Science, Computer Science — CGPA-3.8/4

Guru Nanak Institutions Technical Campus, June 2020 — Hyderabad, India
Bachelor of Technology, Information Technology — CGPA-8.34/10